



Study Notes

Aldehydes

For any queries write to us at hello@ixambee.com

Study notes (Aldehydes)

Carbonyl group

- The organic compounds which contain carbon-oxygen double bond ($>C=O$) are called as carbonyl compounds.
- This is one of the most important functional group in organic chemistry.
- The carbonyl carbon atom is sp^2 hybridised and forms three **sigma (σ)** bonds (two with adjacent hydrogen groups and one with oxygen) and one **pi (Π)** bond (with oxygen atom).

What are Aldehydes?

- In aldehydes, the carbonyl is bonded to a carbon and hydrogen.
- In other words, any of a class of organic compounds in which a carbon atom shares a double bond with an oxygen atom, a single bond with a hydrogen atom, and a single bond with another atom or group of atoms (designated R in general chemical formulas and structure diagrams).
- The general formula for this group is $C_nH_{2n}O$.
- They are represented as-



- For example-
For $n=2$
 $C_nH_{2n}O = C (n)=2; H (2n)=2 \times 2=4$
Hence, the structure of the compound will be C_2H_4O .

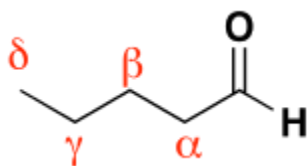
Similarly, for $n=3$:
 $C_nH_{2n}O = C (n)=3; H (2n)=2 \times 3=6$
The compound will be C_3H_6O .

Nomenclature

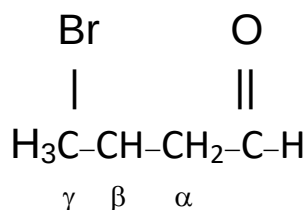
- Aldehydes are often called by their common names instead of their IUPAC name.
- The common names of most aldehydes are derived from the common names of the corresponding carboxylic acids by replacing the ending **-ic** of acid with **aldehyde**.
- The location of the substituent in the carbon chain is indicated by Greek letters $\alpha, \beta, \gamma, \delta$, etc.

Study notes (Aldehydes)

- The atom adjacent to the carbonyl group is α , the next is β and so on.



- For example- β -Bromobutyraldehyde



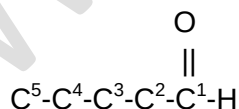
IUPAC Nomenclature

- The IUPAC names of open chain aldehydes are derived from the names of the corresponding alkanes by replacing the **-e** with **-al**.
- Firstly, the longest carbon chain is numbered starting from the carbon of the aldehyde group.
- The substituents are prefixed in alphabetical order along with numerals indicating their positions in the carbon chain.
- Let us take the example of 3-methylpentanal.

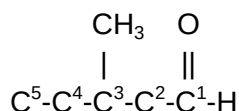
- First draw five carbon atom chain.



- Make aldehyde group at carbon-1

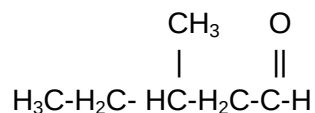


- Attach methyl group at carbon-3 starting from the aldehyde group.



Study notes (Aldehydes)

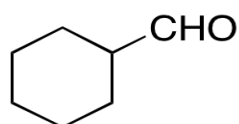
- Now satisfy the valency of each carbon by adding H-group in them.



- Thus, we arrive at the correct structure.

• In case of rings

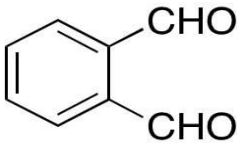
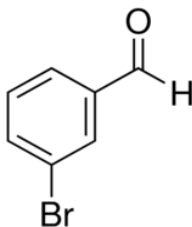
- When the aldehyde group is attached to a ring, the suffix **carbaldehyde** is added after the full name of the cycloalkane.
- The numbering of the ring carbon atoms start from the carbon atom attached to the aldehyde group.
- For example- Cyclohexanecarbaldehyde ($\text{C}_7\text{H}_{12}\text{O}$)



Common and IUPAC names of some aldehydes

Structure	Common name	IUPAC name
HCHO	Formaldehyde	Methanal
CH_3CHO	Acetaldehyde	Ethanal
$(\text{CH}_3)_2\text{CHCHO}$	Isobutyraldehyde	2-Methylpropanal
	γ -Methylcyclohexane	3-Methylcyclohexanecarbaldehyde
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$	Valeraldehyde	Pentanal

Study notes (Aldehydes)

$\text{CH}_2=\text{CHCHO}$	Acrolein	Prop-2-enal
	Phthalaldehyde	Benzene-1,2-dicarbaldehyde
	<i>m</i> -Bromobenzaldehyde	3-Bromobenzenecarbaldehyde Or 3-Bromobenzaldehyde

Physical properties

- Lower members of aldehydes (upto C_{10}) are colourless, volatile liquids **except** formaldehyde, which is gas at room temperature.
- Higher members of aldehydes are **solids** with **fruity odour**.
- Lower aldehydes have **sharp pungent odour**.

Boiling point and solubility

- The boiling of aldehydes are higher than hydrocarbons and ethers of comparable molecular masses due to weak dipole-dipole interactions.
- Their boiling points are lower than alcohols of similar molecular masses due to absence of intermolecular hydrogen-bonding.
- Lower members of aldehydes are **soluble in water**.
- They are fairly soluble in organic solvents like benzene, ether, methanol, chloroform, etc.
- Their solubility decreases rapidly on increasing the length of alkyl chain.

Uses of aldehydes in everyday life

- Aldehydes are used as solvents in chemical industry.
- These are also used as starting materials and reagents for the synthesis of other organic compounds.
- Formalin** (40% solution of formaldehyde) is used to preserve biological specimens.
- Formaldehyde** is also used for the synthesis of Bakelite (phenol-formaldehyde resin), urea-formaldehyde glues and other polymeric products.

Study notes (Aldehydes)

- **Acetaldehyde** (CH_3CHO) is used as starting material for the manufacture of acetic acid, ethyl acetate, vinyl acetate, polymers and drugs.
- **Benzaldehyde** is used in perfumery and in dye industries.
- **Butyraldehyde**, **vanillin** (from vanilla beans), etc. are used as flavoring agents.